Edge Detection Using Laplacian

Shree K. Nayar

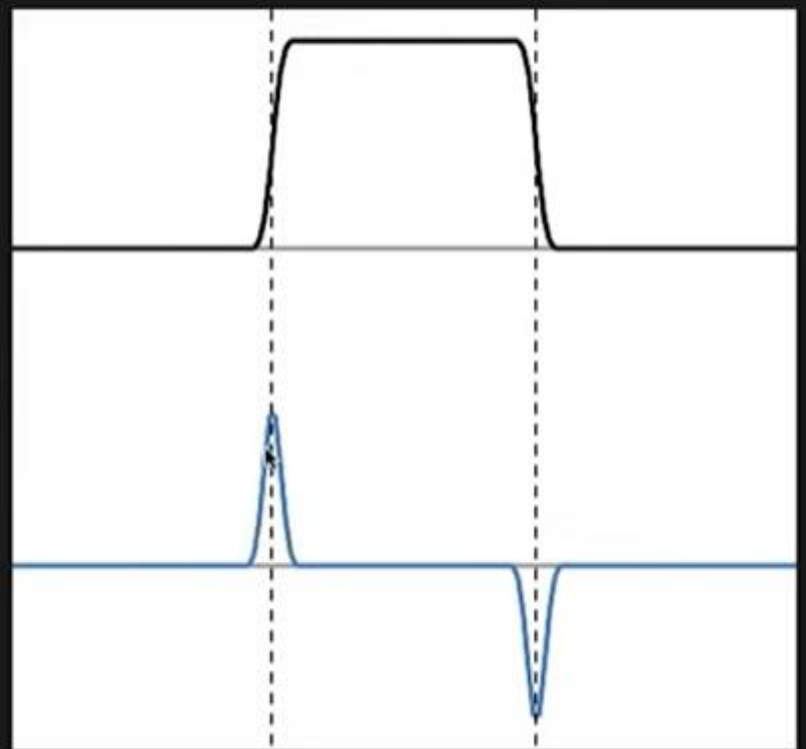
Columbia University

Topic: Edge Detection, Module: Features

First Principles of Computer Vision

© 2020 Shree K. Nayar

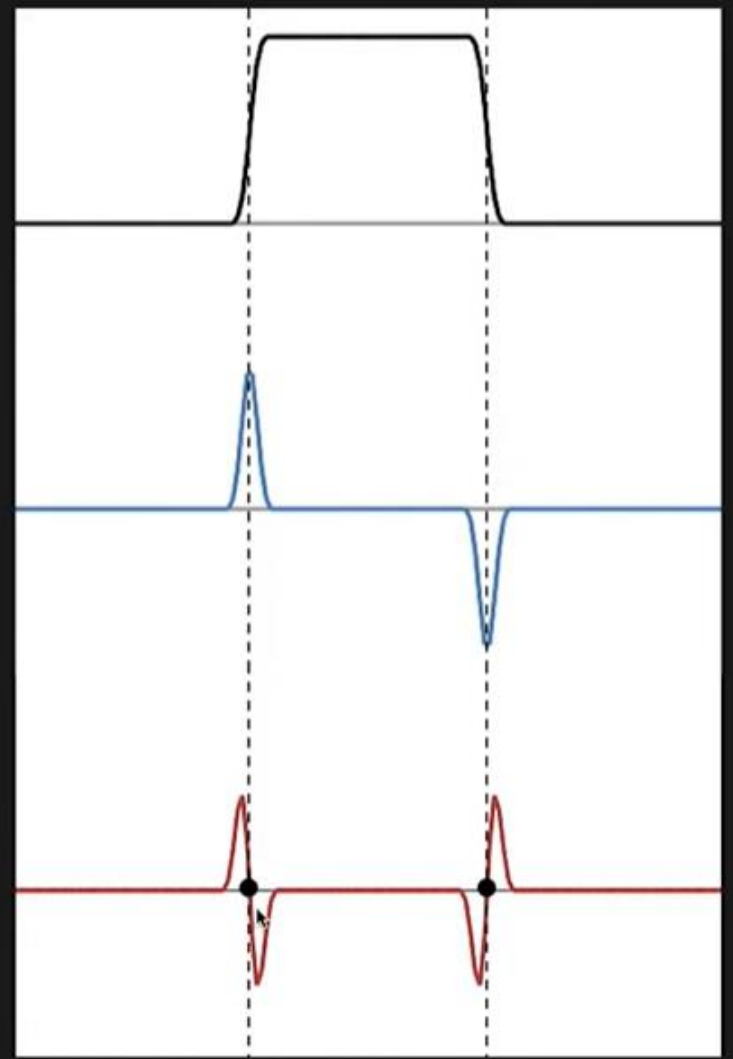
Edge Detection Using 2nd Derivative



First Derivative: $\frac{\partial f}{\partial x}$

Local Extrema
Indicate Edges

Edge Detection Using 2nd Derivative



First Derivative: $\frac{\partial f}{\partial x}$

Second Derivative: $\frac{\partial^2 f}{\partial x^2}$

Local Extrema
Indicate Edges

Zero-Crossings
Indicate Edges

Laplacian (∇^2) as Edge Detector

Laplacian: Sum of Pure Second Derivatives

$$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$$

Pronounced as "Del Square I"

- Edges are "zero-crossings" in Laplacian of image
- Laplacian does not provide directions of edges

Discrete Laplacian(∇^2) Operator

Finite difference approximations:

$$\frac{\partial^2 I}{\partial x^2} \approx \frac{1}{\varepsilon^2}\left(I_{i-1,j} - 2I_{i,j} + I_{i+1,j}\right)$$

$$\frac{\partial^2 I}{\partial y^2} \approx \frac{1}{\varepsilon^2}\left(I_{i,j-1} - 2I_{i,j} + I_{i,j+1}\right)$$

$$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$$

$I_{i-1,j+1}$	$I_{i,j+1}$	$I_{i+1,j+1}$
$I_{i-1,j}$	$I_{i,j}$	$I_{i+1,j}$
$I_{i-1,j-1}$	$I_{i,j-1}$	$I_{i+1,j-1}$

ε

Convolution Mask:

$$\nabla^2 \approx \frac{1}{\varepsilon^2}\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{OR} \quad \nabla^2 \approx \frac{1}{6\varepsilon^2}\begin{bmatrix} 1 & 4 & 1 \\ 4 & -20 & 4 \\ 1 & 4 & 1 \end{bmatrix}$$

(More Accurate

© 2020 Shree K. Nayar

Laplacian Edge Detector



Image (I)



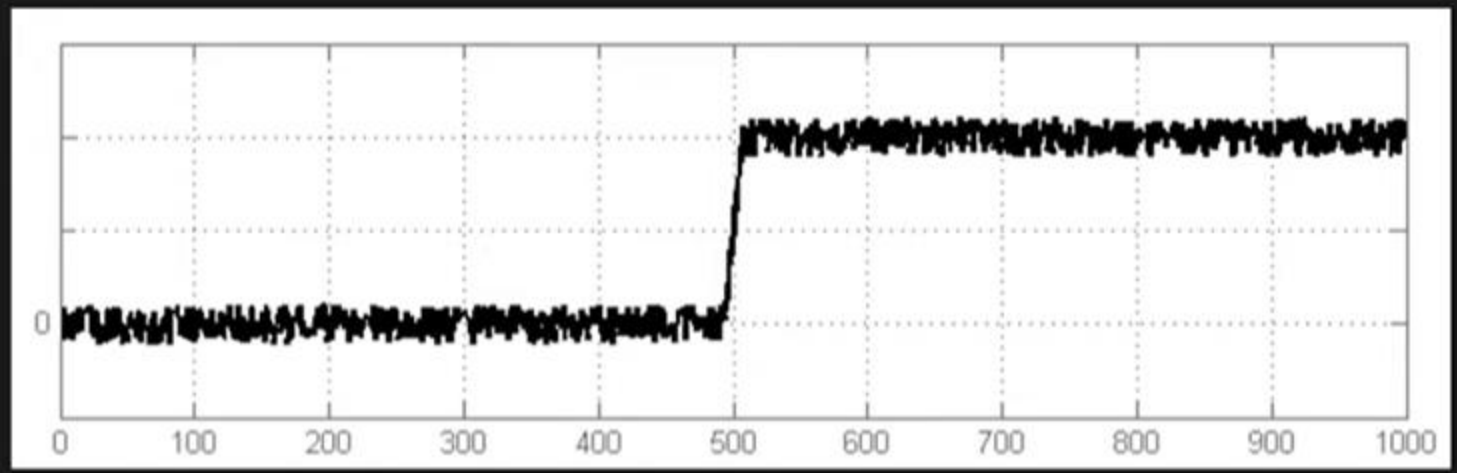
Laplacian
(0 maps to 128)



Laplacian
"Zero Crossings"

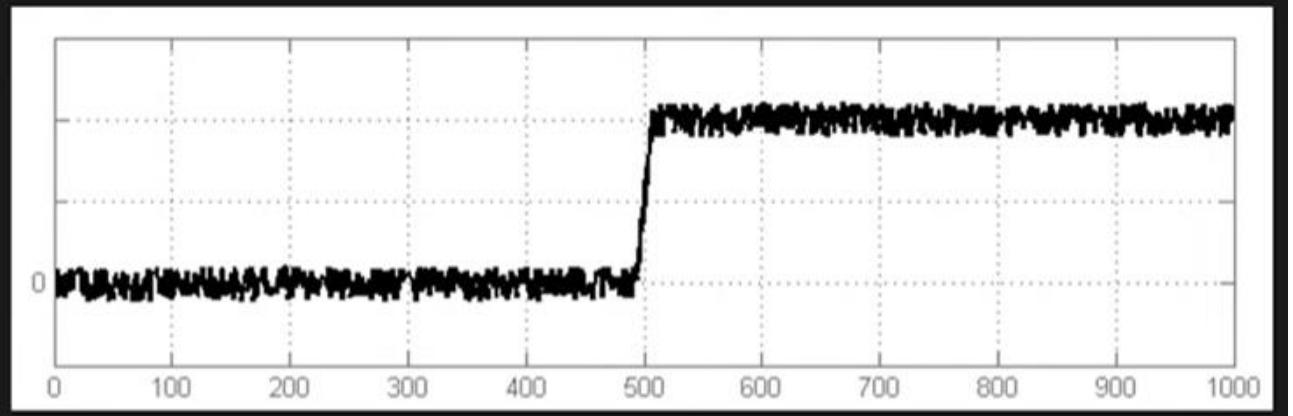
Effects of Noise

$f(x)$



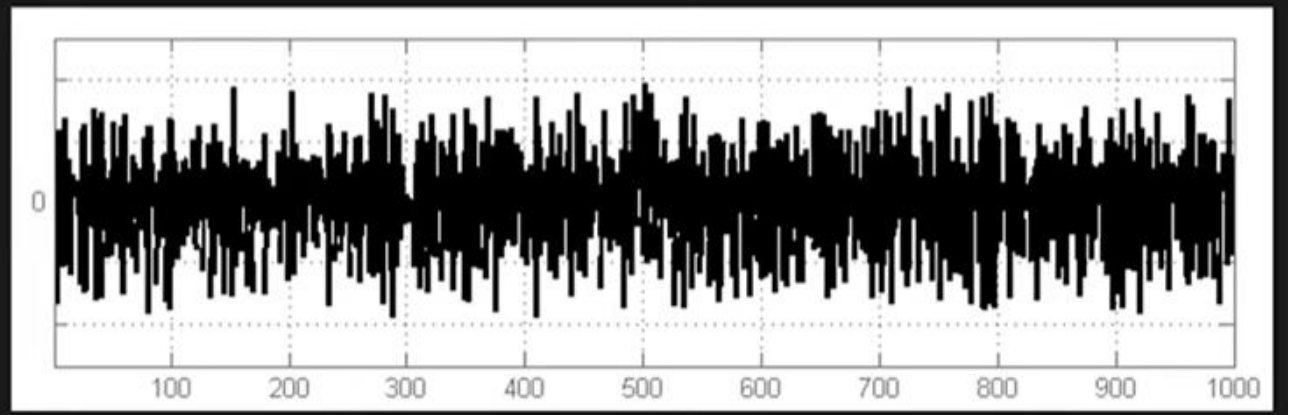
Effects of Noise

$f(x)$



$\nabla f(x)$

(Gradient)



Where is the edge??

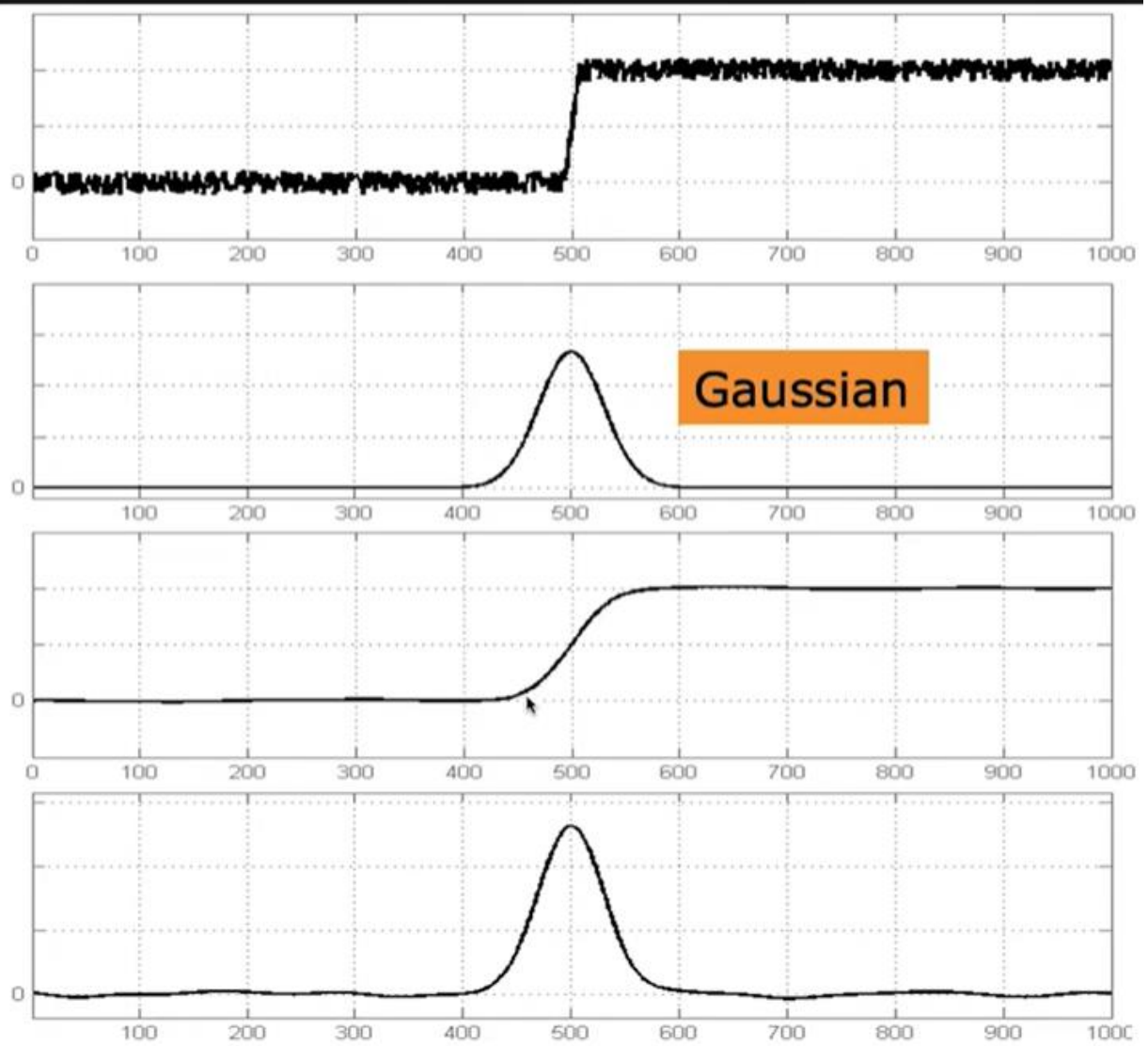
Solution: Gaussian Smooth First

f

n_σ

$n_\sigma * f$

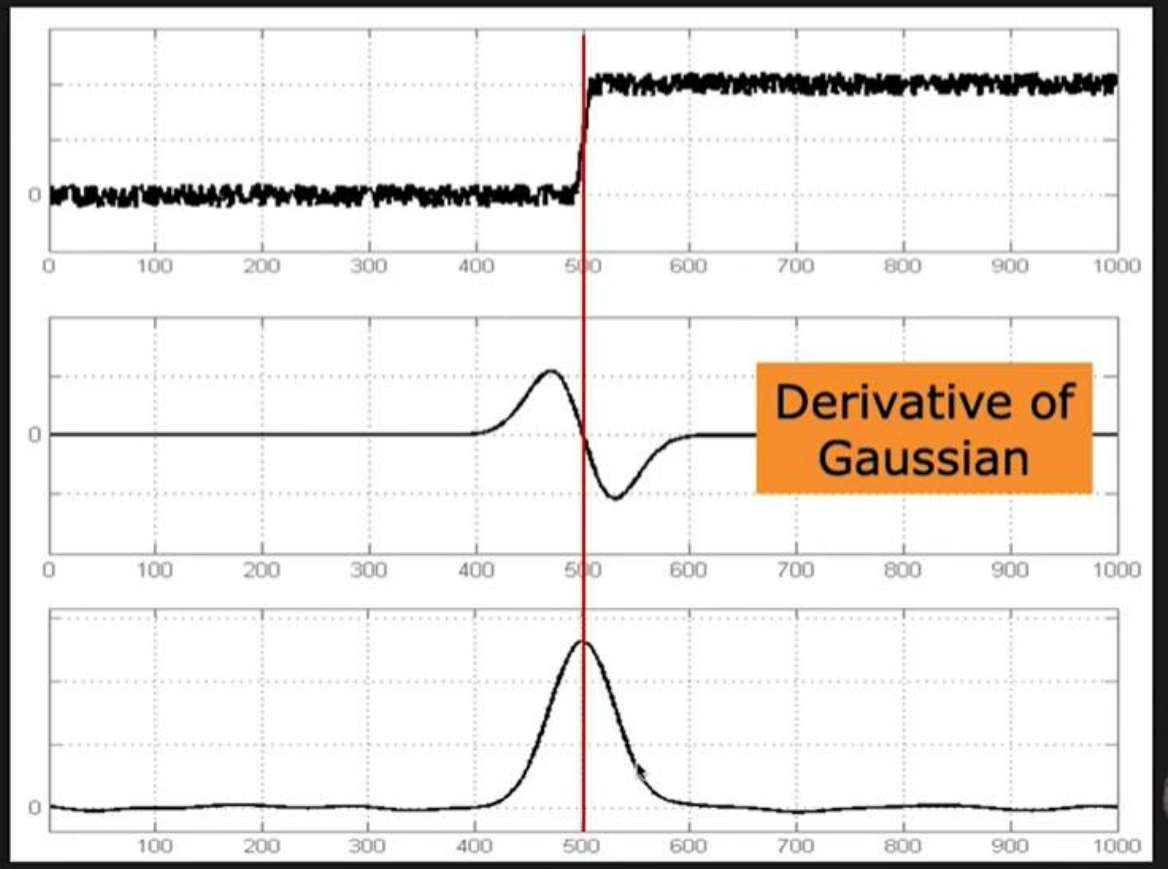
$\nabla(n_\sigma * f)$



20 Shree K. Nayar

Derivative of Gaussian ($\nabla(n_\sigma)$)

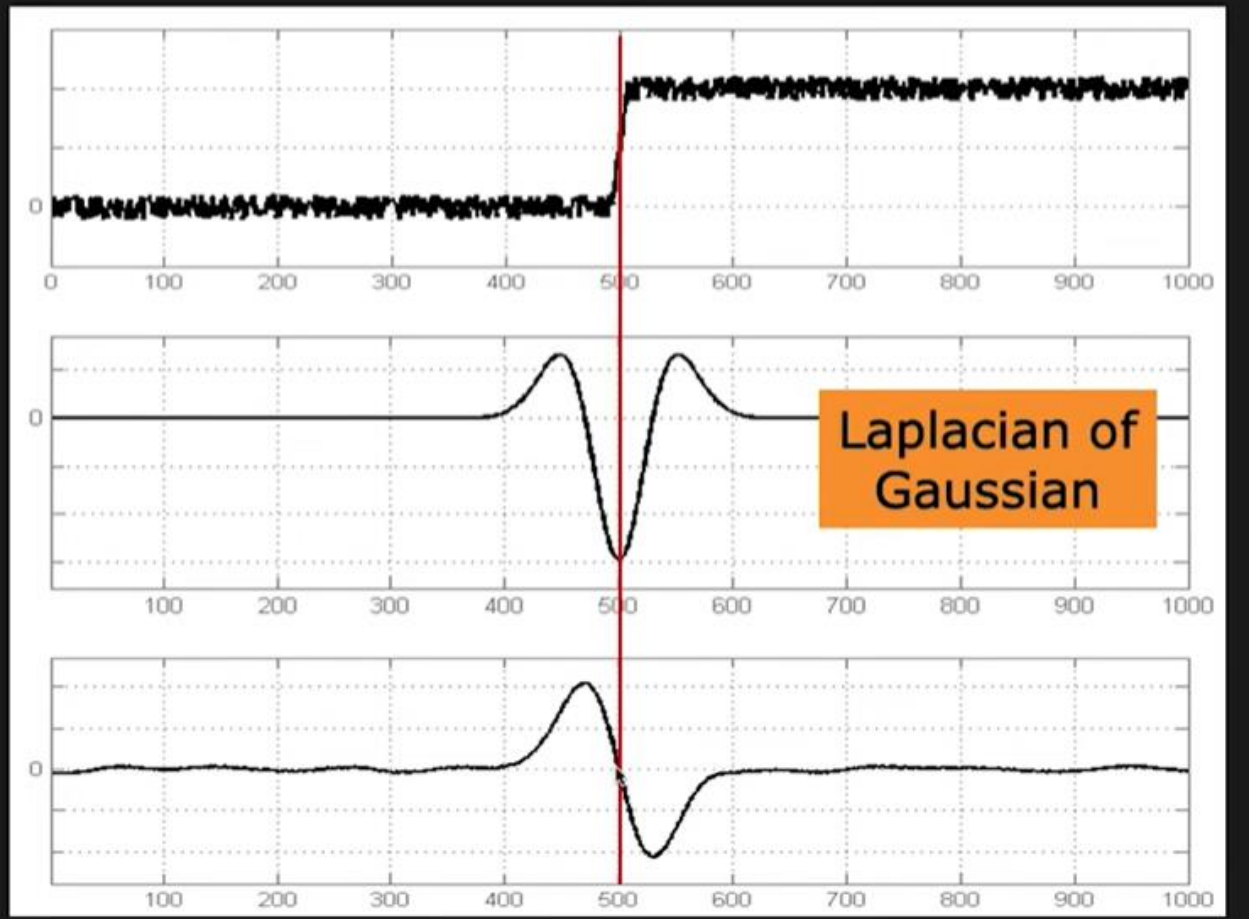
$\nabla(n_\sigma * f) = \nabla(n_\sigma) * f$ ...saves us one operation.



© 2020 Shree K. Nayar

Laplacian of Gaussian ($\nabla^2 n_\sigma$ or $\nabla^2 G$)

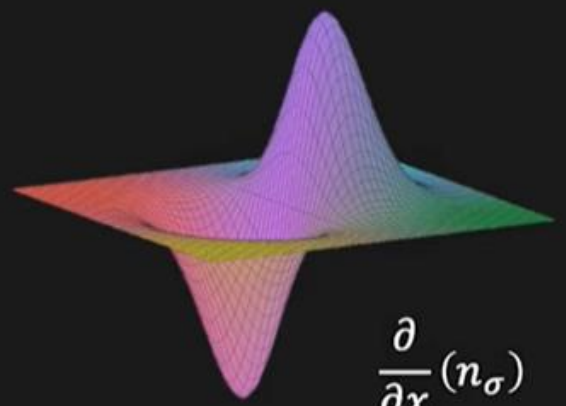
$$\nabla^2(n_\sigma * f) = \nabla^2(n_\sigma) * f$$ ...saves us one operation.



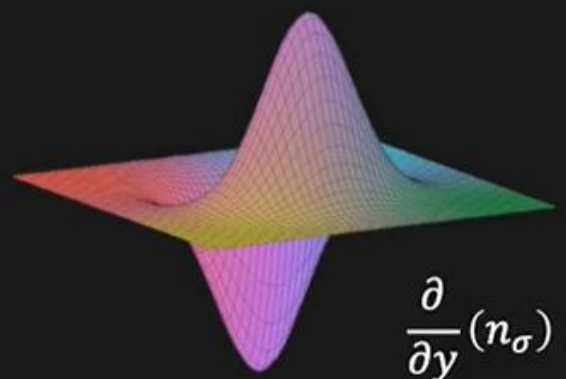
2020 Shree K. Nayar

Gradient vs. Laplacian

Derivative of Gaussian (∇G)

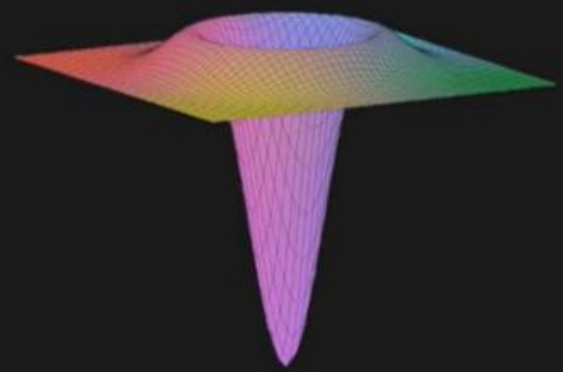


$$\frac{\partial}{\partial x}(n_\sigma)$$



$$\frac{\partial}{\partial y}(n_\sigma)$$

Laplacian of Gaussian ($\nabla^2 G$)



$$\frac{\partial^2}{\partial x^2}(n_\sigma) + \frac{\partial^2}{\partial y^2}(n_\sigma)$$

Inverted "Sombrero"
(Mexican Hat)

© 2020 Shree K. Nayar

Gradient vs. Laplacian

Provides location, magnitude and direction of the edge.	Provides only location of the edge.
Detection using Maxima Thresholding.	Detection based on Zero-Crossing.
Non-linear operation. Requires two convolutions.	Linear Operation. Requires only one convolution.

An operator that has the best of both?

© 2020 Shree K. Nayar

Canny Edge Detector

Shree K. Nayar

Columbia University

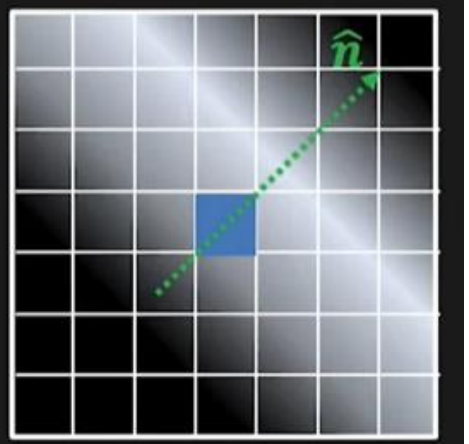
Topic: Edge Detection, Module: Features

First Principles of Computer Vision

Canny Edge Detector

- Smooth Image with 2D Gaussian: $n_\sigma * I$
- Compute Image Gradient using Sobel Operator: $\nabla n_\sigma * I$
- Find Gradient Magnitude at each pixel: $\|\nabla n_\sigma * I\|$
- Find Gradient Orientation at each Pixel:

$$\hat{\boldsymbol{n}} = \frac{\nabla n_\sigma * I}{\|\nabla n_\sigma * I\|}$$



$\|\nabla n_\sigma * I\|$

© 2020 Shree K. Nayar

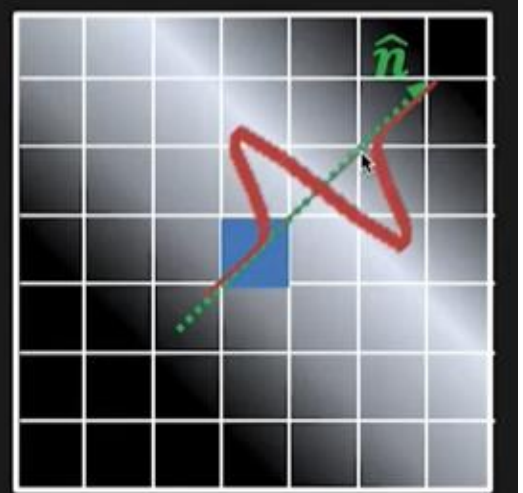
Canny Edge Detector

- Smooth Image with 2D Gaussian: $n_\sigma * I$
- Compute Image Gradient using Sobel Operator: $\nabla n_\sigma * I$
- Find Gradient Magnitude at each pixel: $\|\nabla n_\sigma * I\|$
- Find Gradient Orientation at each Pixel:

$$\hat{\boldsymbol{n}} = \frac{\nabla n_\sigma * I}{\|\nabla n_\sigma * I\|}$$

- Compute Laplacian along the Gradient Direction $\hat{\boldsymbol{n}}$ at each pixel

$$\frac{\partial^2 (n_\sigma * I)}{\partial \hat{\boldsymbol{n}}^2}$$



$\|\nabla n_\sigma * I\|$

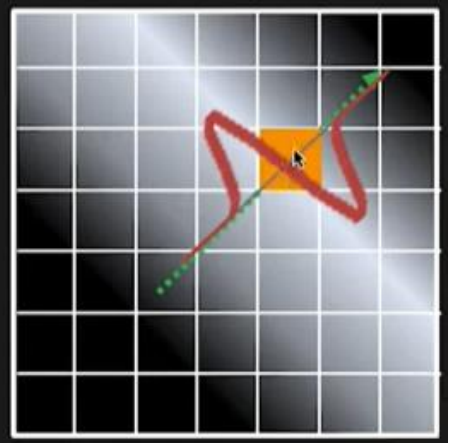
Canny Edge Detector

- Smooth Image with 2D Gaussian: $n_\sigma * I$
- Compute Image Gradient using Sobel Operator: $\nabla n_\sigma * I$
- Find Gradient Magnitude at each pixel: $\|\nabla n_\sigma * I\|$
- Find Gradient Orientation at each Pixel:

$$\hat{\boldsymbol{n}} = \frac{\nabla n_\sigma * I}{\|\nabla n_\sigma * I\|}$$

- Compute Laplacian along the Gradient Direction $\hat{\boldsymbol{n}}$ at each pixel

$$\frac{\partial^2 (n_\sigma * I)}{\partial \hat{\boldsymbol{n}}^2}$$



$\|\nabla n_\sigma * I\|$

- Find Zero Crossings in Laplacian to find the edge location

© 2020 Shree K. Nayar

Canny Edge Detector

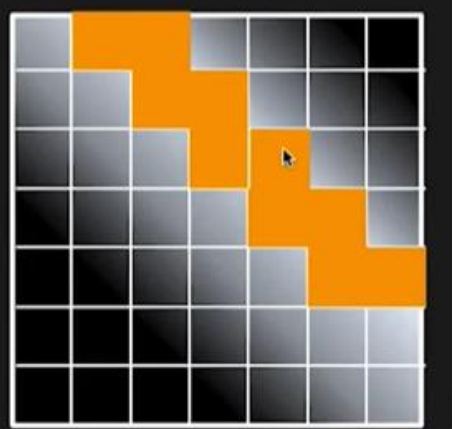
- Smooth Image with 2D Gaussian: $n_\sigma * I$
- Compute Image Gradient using Sobel Operator: $\nabla n_\sigma * I$
- Find Gradient Magnitude at each pixel: $\|\nabla n_\sigma * I\|$
- Find Gradient Orientation at each Pixel:

$$\hat{\boldsymbol{n}} = \frac{\nabla n_\sigma * I}{\|\nabla n_\sigma * I\|}$$

- Compute Laplacian along the Gradient Direction $\hat{\boldsymbol{n}}$ at each pixel

$$\frac{\partial^2 (n_\sigma * I)}{\partial \hat{\boldsymbol{n}}^2}$$

- Find Zero Crossings in Laplacian to find the edge location



$|\nabla n_\sigma * I|$

© 2020 Shree K. Nayar

Canny Edge Detector Results



Image



$\sigma = 1$



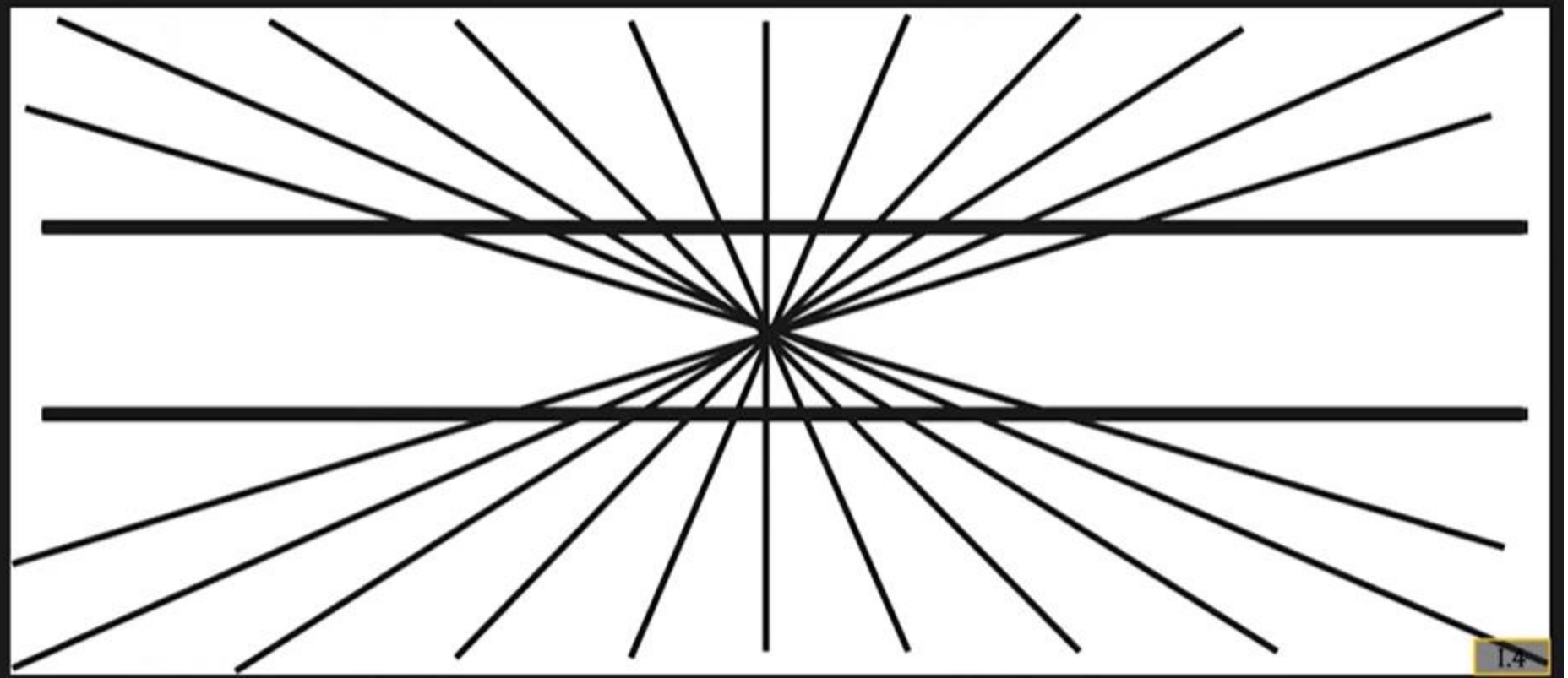
$\sigma = 2$



$\sigma = 4$

© 2020 Shree K. Nayar

Edge Illusions: Hering Illusion



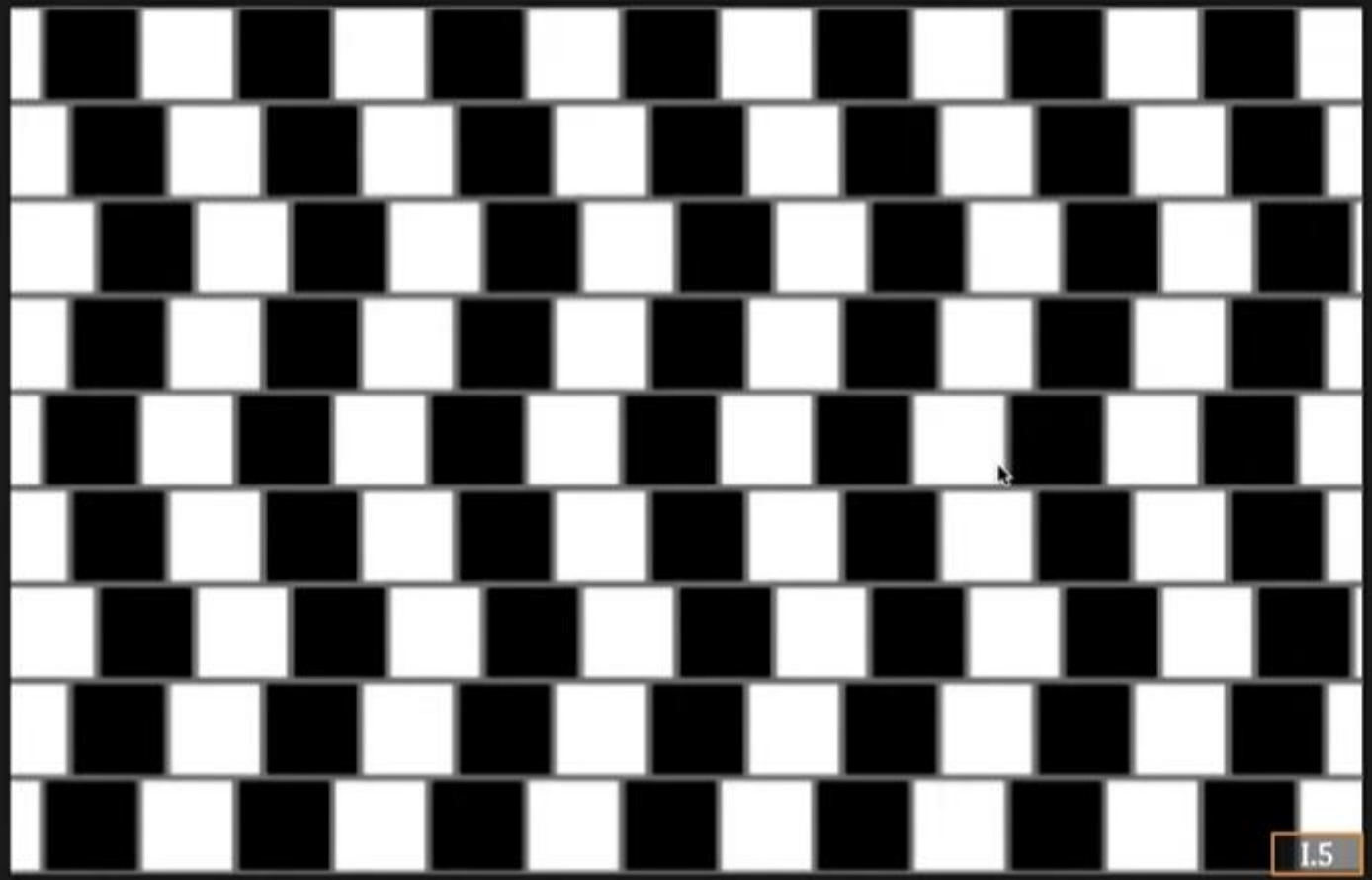
Ewald Hering, 1861

Edge Illusions: Hering Illusion



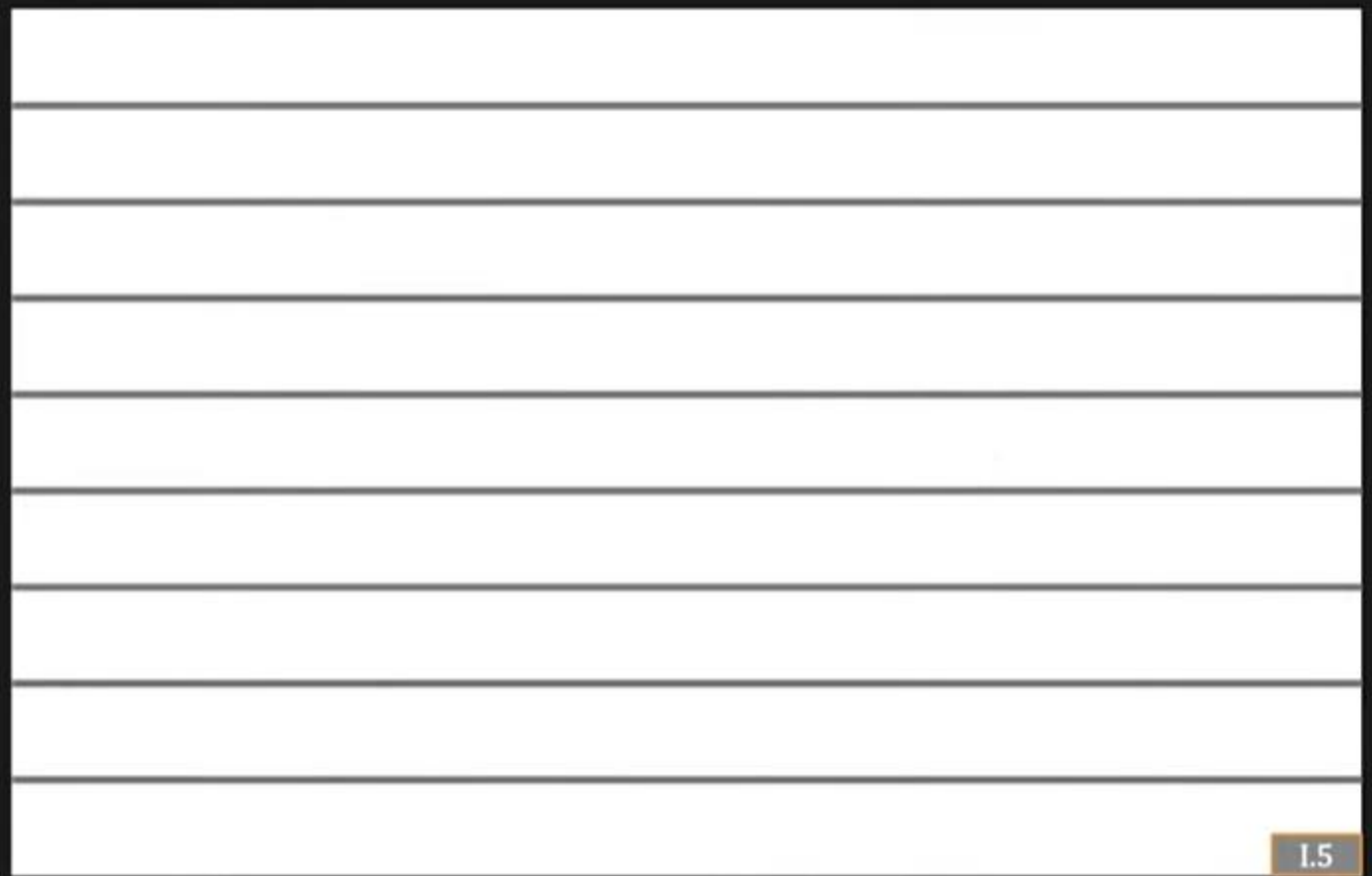
Ewald Hering, 1861

Edge Illusions: Café Wall Illusion



Gregory and Heard, 1979

Edge Illusions: Café Wall Illusion



Gregory and Heard, 1979